----- Active space -----

INFLAMMATORY BOWEL DISEASE : PART 1

Entities Of Inflammatory Bowel Disease

00:03:45

1. Ulcerative colitis (UC).
2. Crohn's disease (CD).
3. Microscopic colitis.
4. Diversion colitis.

Currently, inflammatory bowel disease includes only UC and CD.

MICROSCOPIC COLITIS

Associated with long-term use of PPI and NSAIDS.

Features :

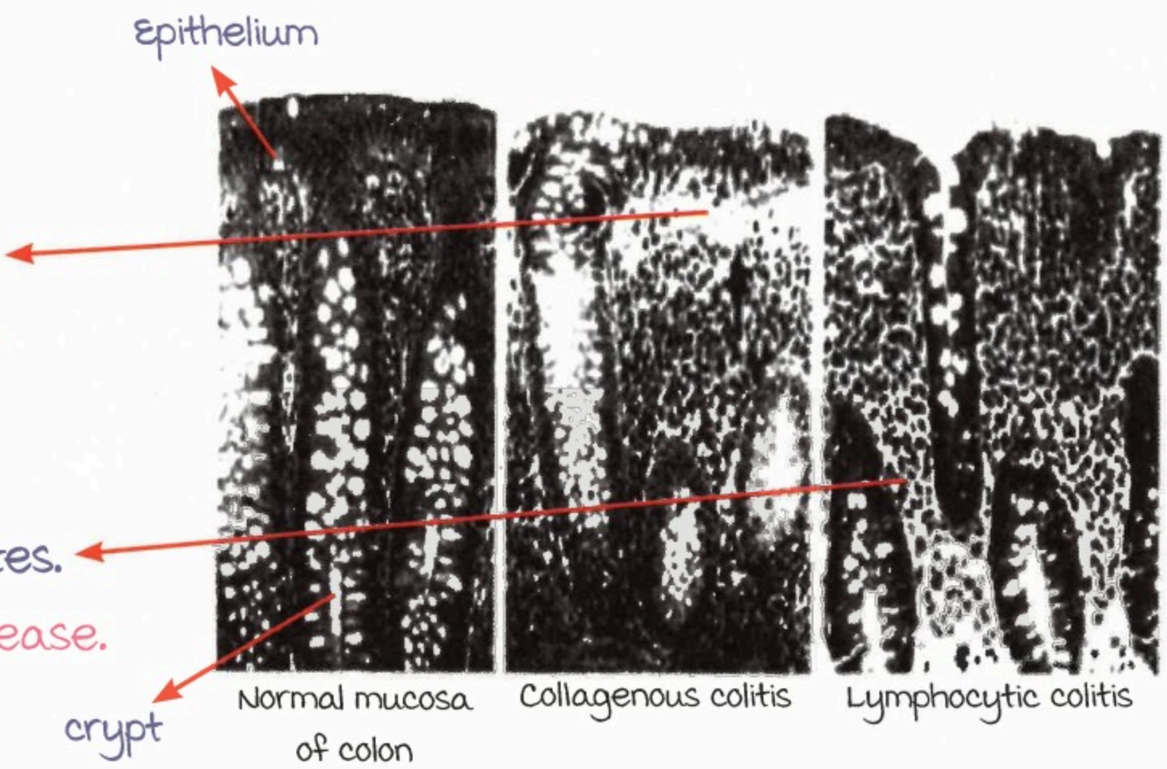
- Presents with chronic, non-bloody, voluminous, watery diarrhea (Similar to small bowel diarrhea).
- On clinical examination : Normal.

Investigations :

Endoscopy : Normal.

Biopsy : Two patterns

1. Collagenous Colitis :
   - Collagen band in mucosa.
   - Associated with Sjogren's syndrome.
2. Lymphocytic colitis :
   - Intra-epithelial lymphocytes.
   - Associated with celiac disease.



DIVERSION COLITIS

Colitis seen in patients using colostomy bags (After Sx for bowel malignancies).

NORMAL HISTOLOGY OF COLON

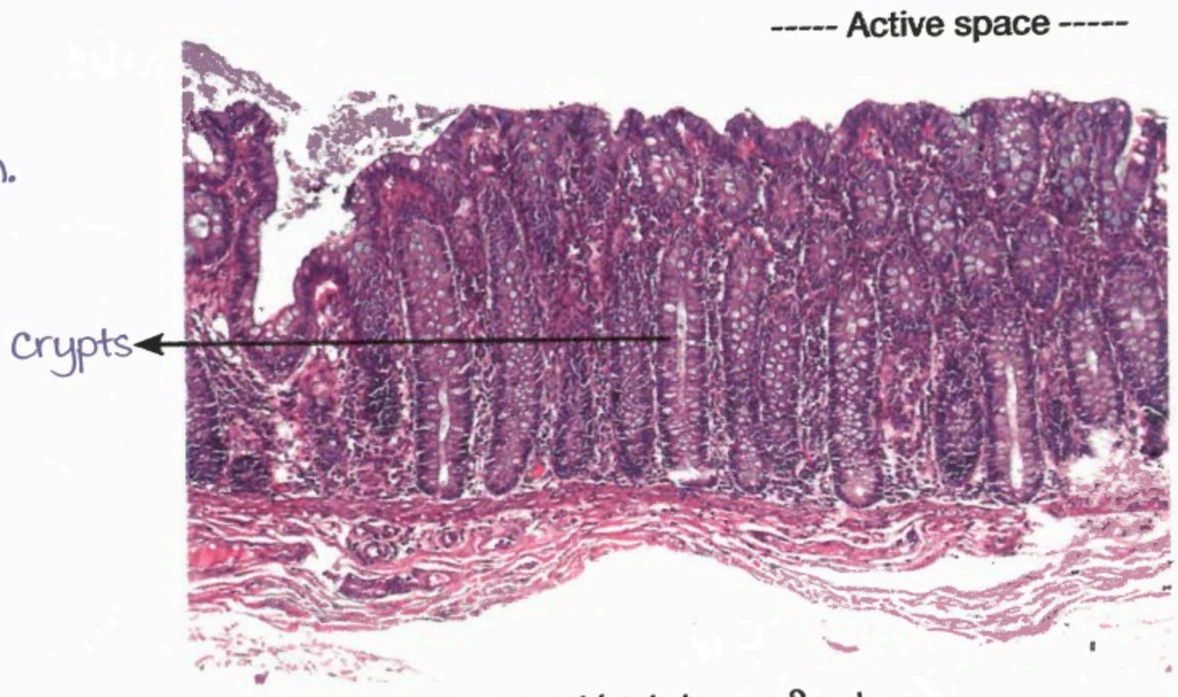
Layers : (Innermost to outermost).

- Mucosal epithelial cells.
- Lamina propria.
- Muscularis mucosa.
- Submucosa.
- Muscularis propria.
- Serosa.

Mucosa :

- Villi are absent.
- Crypts : Contain microvilli → water absorption.
- Goblet cells : Secrete mucus.

Note: Diarrhea → >200 ml of stool volume.



Normal histology of colon

Ulcerative Colitis And Crohn's Disease

00:10:25

EPIDEMIOLOGY

Overall :

- CD > UC.
- Males > Females.

	UC	CD
Female : male ratio	Males > Females	Females > males (OCP use)
Age	2 peaks : 15–30 years & 70–90 years.	
Ethnicity	Jews > Non-Jewish white > African American > Hispanic > Asian	
Smoking	Protective	Causative
Appendicectomy	Protective	No association
OCP	No increased risk	Increased risk
Familial Inheritance (Concordance)	Significant CD > UC	
Antibiotics in 1st year of life	Increases risk of IBD by 3 times	

Note :

- Behcet's disease :
  - GI manifestations like Crohn's disease.
  - Share a common antibody : ASCA (Anti-Saccharomyces cerevisiae antibody).
- Prolonged use of α-methyl dopa causes features similar to ulcerative colitis.
- Campylobacter jejuni is a risk factor for IBD (Also causes large bowel diarrhea, Guillain Barre syndrome).

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Genetic Syndromes Associated With IBD

00:18:24

Present as IBD in childhood :

UC (Non-granulomatous)	CD (Granulomatous)
Turner's syndrome : m/c	
Wiskott-Aldrich Syndrome	• Von Gierke's disease. • Hermansky-Pudlak syndrome.

CLINICAL FEATURES

Syndrome	Defect/Inheritance	Features
Wiskott-Aldrich Syndrome.	WASP gene mutation (X-linked recessive)	Triad of: 1. Recurrent sino-pulmonary infections. 2. Recurrent eczema. 3. Recurrent thrombocytopenia (micro platelets).
Hermansky-Pudlak Syndrome.	Autosomal recessive	• Platelet dysfunction with normal platelet count. • Interstitial lung disease : Pulmonary fibrosis. • Oculocutaneous albinism.
Von Gierke's Disease (Glycogen storage disorder - type I).	Glucose-6-phosphatase deficiency	• Hypoglycemia +/- Seizures, Doll facies. • Hepatomegaly. • Lactic acidosis. • Hyperuricemia. • Hyperlipidemia.

IMMUNE POLYENDOCRINOPATHY X-LINKED SYNDROME (IPEX)

Pathophysiology :

Physiology :

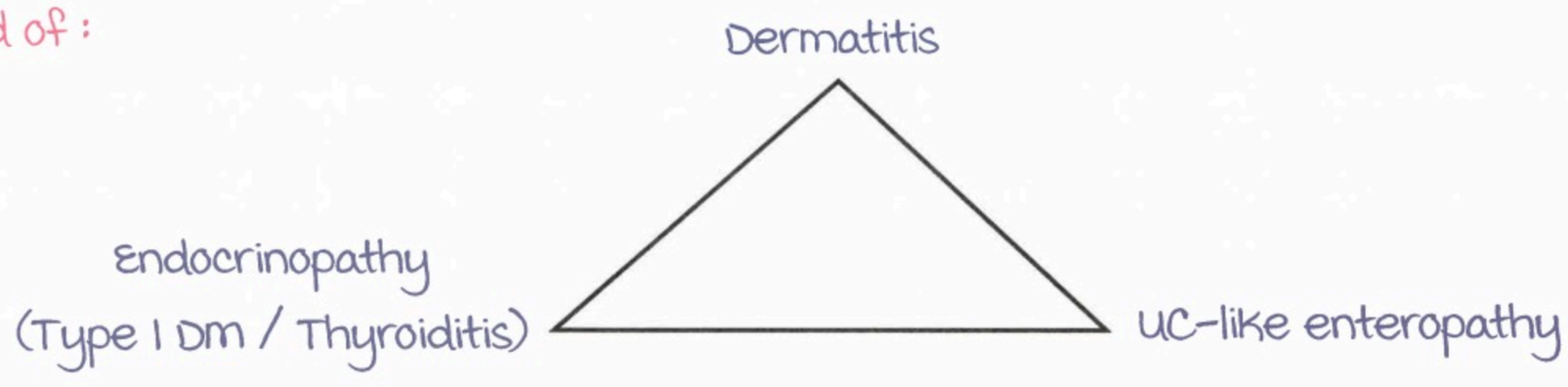
FOXP3 : Transcription factor → Normal T-reg cell function (Cell tolerance).

Pathology :

Loss of FOXP3 function → Reduced cell tolerance → IPEX syndrome.

Clinical Features :

Triad of :



Management :

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- Supportive care and replacement therapy.
- Gene therapy.

Investigations : Genetic testing.

Note :

Markers of T-reg cell :

- CD3 : Pan T-cell marker
- CD4 : Helper T-cell.
- CD25 : T-reg cell.

EARLY ONSET CHILDHOOD IBD

- Defect in IL-10 or IL-10 receptor function.
- Severe refractory IBD in early life.

Note:

most potent anti-inflammatory cytokine in the body : IL-10.

Pathogenesis Of IBD

00:26:15

Exogenous and endogenous factors → Dysregulated mucosal immune response in a genetically susceptible individual.

ULCERATIVE COLITIS

- Pathogenesis is unclear.
- Overactive adaptive immune system : Th_2 lineage.

CROHN'S DISEASE

1. Genetically susceptible individuals :

   NOD 2 (CARD 15) gene polymorphism on chromosome 16.

   ↓

   Dysregulated mucosal immune response :

   Defective innate immunity :

   - Decreased clearance of intra-cellular organisms (mycobacterium paratuberculosis, M. avium complex).
   - Decreased defensin production from Paneth cells.
   - Defective autophagy.

   ↓

   Overactive adaptive immune system (Th_1 lineage : Crohn's disease).

2. Decrease in gut bacteria like Faecalibacterium (Gut protective).

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Pathology : Ulcerative Colitis

00:33:00

SITE OF INVOLVEMENT

- Rectum alone : 50%.
- Extension to sigmoid colon : 30–40%.
- Pancolitis (Rectum, sigmoid, colon involved) : 10–20%.
  10–20% of pancolitis patients : Ileum involvement → Backwash ileitis.

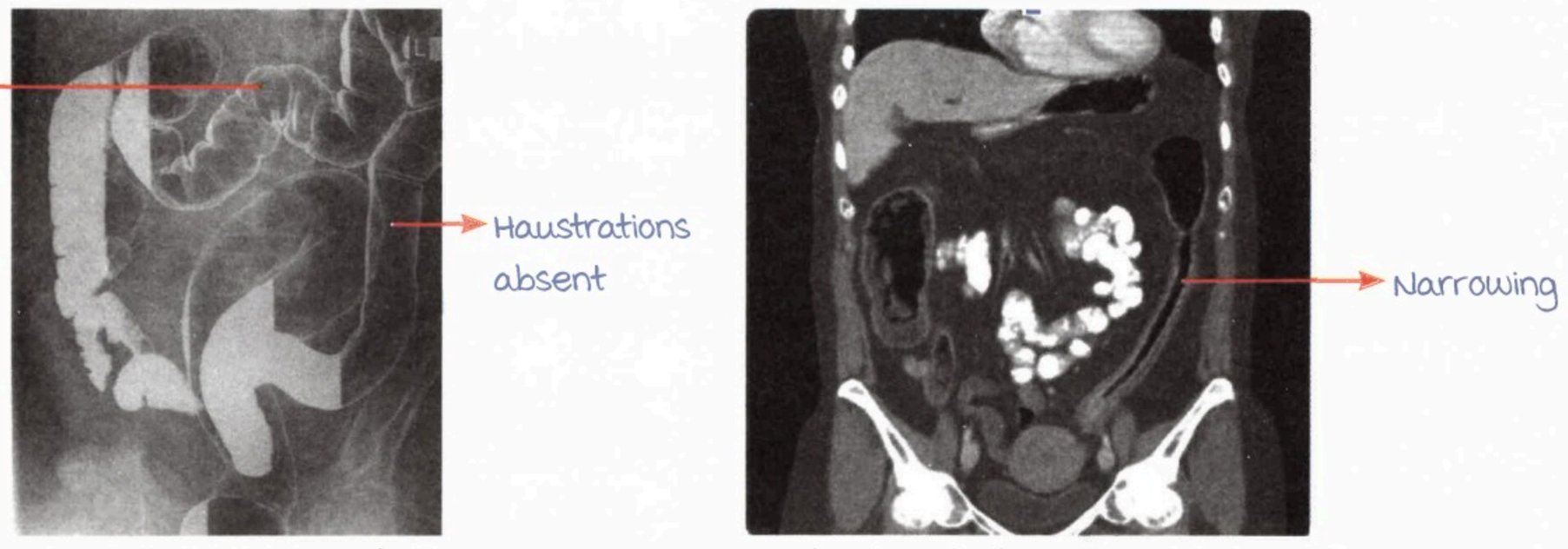
MACROSCOPY

- Superficial, continuous, symmetrical lesions :
- Continuous involvement : Rectum → Sigmoid → Colon → Ileum (No skip lesions).
- 1st change : Erythematous mucosa with fine, granular surface
  ↓
  Hemorrhagic, edematous, ulcerated mucosa
  ↓
  Pseudopolyps : Inflammatory polyp d/t epithelial regeneration



Diffuse continuous lesions

- Chronic long-standing UC (rare).
  Lead pipe appearance (Atrophic, featureless colon) :
  a. Loss of haustrations.
  b. Shortening and narrowing of colon.
- Fulminant colitis (very rare) : Toxic megacolon (Colon diameter >6cm).

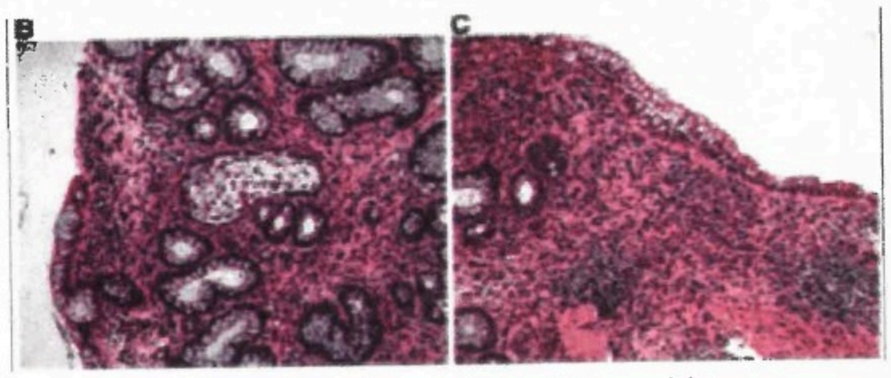


Lead pipe appearance seen on barium study

MICROSCOPY

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- Superficial Involvement : mucosa & superficial part of submucosa.
- Cryptitis with loss of crypt architecture.
- Paneth cell metaplasia (Seen distal to hepatic flexure).
- Basal plasma cell and lymphoid aggregates.
  - Earliest finding.
  - Can help in diagnosing relapse in remission patients.



Chronic active ulcerative colitis

Note :

Factors against the diagnosis of ulcerative colitis :

- Transmural involvement.
- Skip lesions.
- Deep ulcers.
- Thickening.
- Strictures.
- Fissures.
- Fistulas.
- Fibrosis.
- Abscess.

Pathology : Crohn's Disease

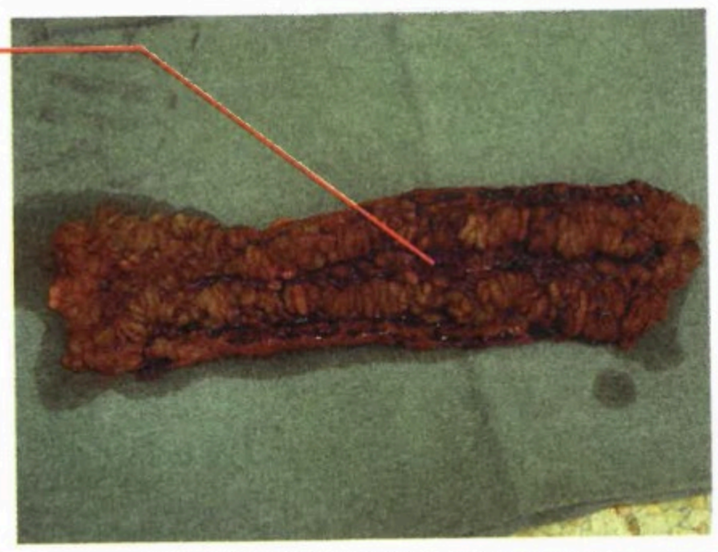
00:44:20

SITE OF INVOLVEMENT

- Ileocolitis : 40–50%.
- Ileitis : 30–40%.
- Colitis : 15–20%.
- Can occur anywhere from oral cavity to anal canal (E.g : Perianal involvement).

MACROSCOPY

- Deep, transmural, asymmetric skip lesions.
- Earliest finding : Aphthous ulcers → Small, translucent ulcers.
- Most visible finding : Linear, Serpenginous ulcers.
- Cobblestone appearance : The ulcers fuse longitudinally and transversely with areas of uninvolved mucosa in between.
- Peri rectal fissures, fistulas and abscesses.



Cobblestone appearance

↓

Fistula tracts resolve by fibrosis

↓

a. Thickening of the wall
b. Strictures

↓

Garden hose appearance → Tubular lumen with extensive transmural fibrosis and stenosis seen in terminal ileum.

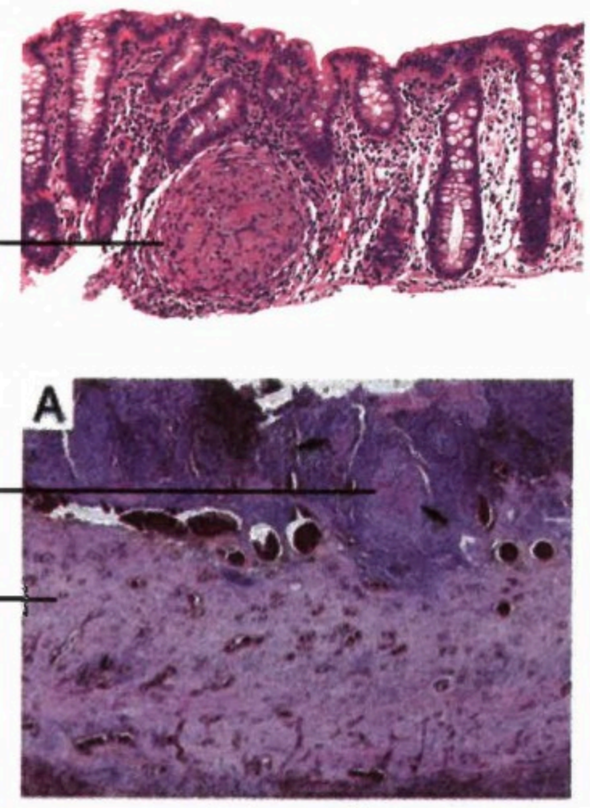
- Creeping fat : Thickened mesentery encasing the bowel wall.

Note : Garden hose appearance in esophagus → Well developed progressive systemic sclerosis.

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MICROSCOPY

- Transmural
  Involvement upto muscularis propria.
  (Serosa involvement : Leads to fissures,fistula etc).
- Non-caseating granulomas
  At any level, can be extraintestinal.
- Mesenteric lymphadenopathy.
- Changes in crypts :
  - Cryptitis.
  - Crypt abscesses.
  - Crypt architecture : Normal.
- Pyloric metaplasia.
- Lymphoid aggregates.



Histology of CD

Note :

Causes of non-caseating granuloma :

- Sarcoidosis.
- Tuberculosis.
- Cat Scratch disease.
- Berylliosis.
- Hypersensitivity pneumonitis.
- Crohn's disease.

INFLAMMATORY BOWEL DISEASE : PART 2

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Ulcerative colitis

00:00:20

CLINICAL FEATURES

- Based on the level of involvement :

Proctitis	Proctosigmoiditis	Proctocolitis
• Painless bleeding PR (m/c) • D/d : a. Hemorrhoids b. Malignancies	Streaks of blood onto formed stool.	Large bowel diarrhea : 1. Small, frequent stools with blood, pus and mucus. 2. Tenesmus
Slow proximal transit → Constipation		-

- Normal physical examination.
- Systemic symptoms : very rare (Th_2 lineage → Involves IL-4, 5, 13).
- 90% have relapsing course.

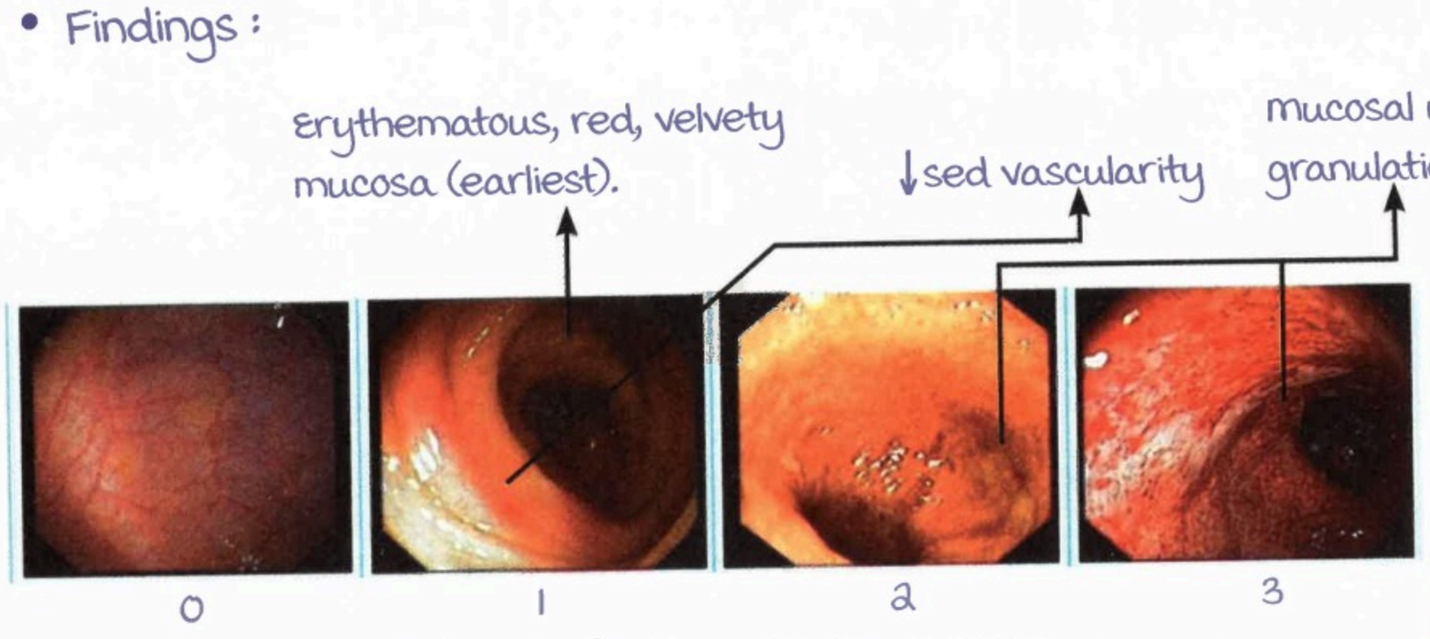
Note :

Fecal calprotectin :

- most sensitive biomarker.
- used to assess disease severity, response to Rx & to diagnose relapse.

ENDOSCOPY

- Symmetrical, superficial, and continuous involvement.
- Findings :



Endoscopy findings with mayo score



Inflammatory pseudo polyps

- Lead pipe/pipestem colon (very rare in endoscopy).

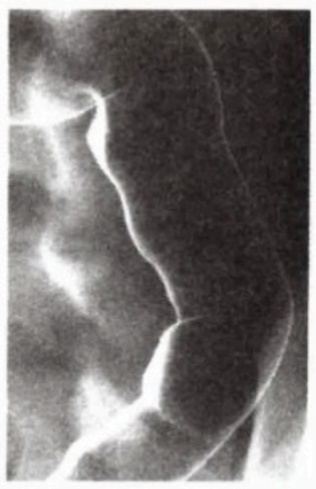
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RADIOLOGY

Barium enema :



Normal mucosa



Fine mucosal granularities



Superficial ulcerations

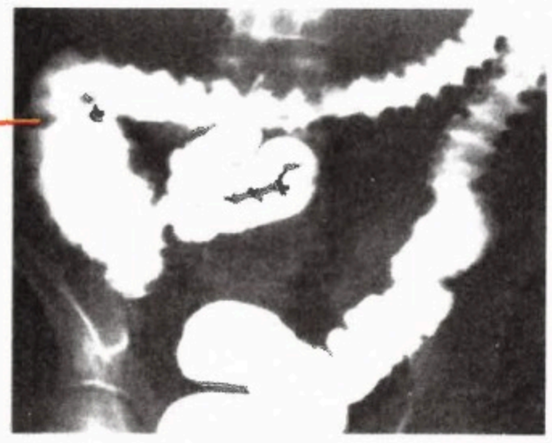


Extensive ulcerations



Inflammatory pseudopolyps

Deep ulcers extending to submucosa



Collar stud ulcers



Lead pipe appearance



Toxic megacolon

- Earliest change : Fine, mucosal granularities.
- Serrations producing hazy margin.
- Mild narrowing of lumen.
- Loss of rectal valves.
- Backwash ileitis (Inflammation of terminal ileum).

Crohn's disease

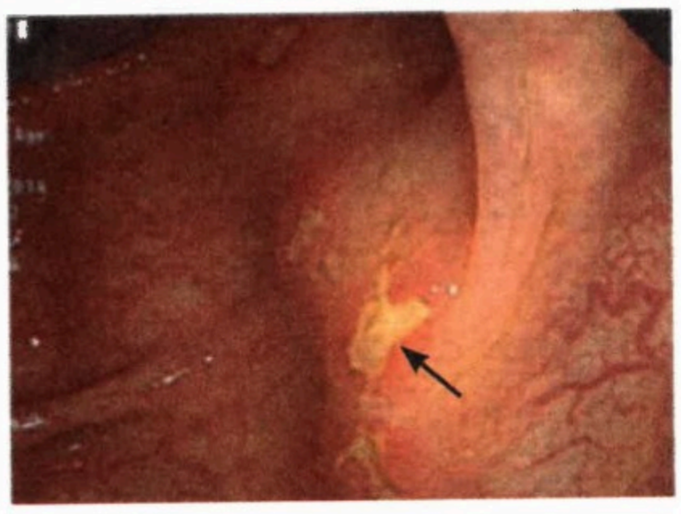
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CLINICAL FEATURES

- Ileocolitis (m/c) :
  - a. Episodic right lower quadrant pain.
  - b. Relieved with diarrhea (Large bowel diarrhea d/t colitis).
- D/t Th_1 profile (IL-2, IFN γ) :
  - a. Fever.
  - b. Weight loss.
  - c. Systemic symptoms.
- Jejunoileitis → malabsorption.
- Palpable inflammatory mass : D/t thickening of bowel wall.
- Bowel obstruction.
- Fistulas : Enterovesical > Perianal.
- Fissures → Fibroses & strictures.

ENDOSCOPY

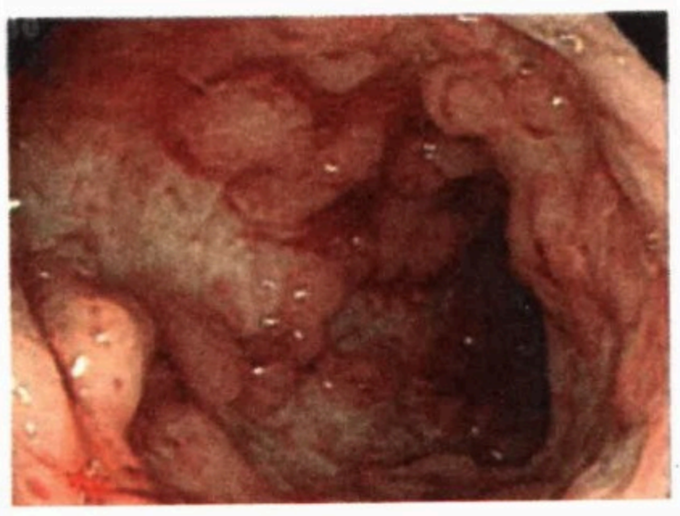
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Aphthous ulcer



Longitudinal ulceration



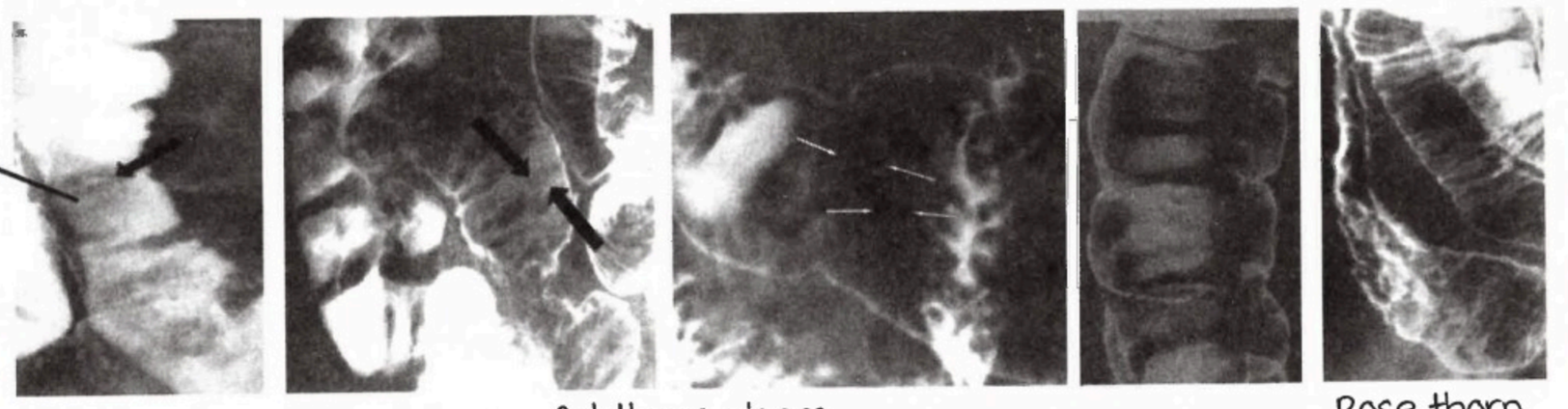
Cobblestone appearance

RADIOLOGY

Barium enema :

Target Sign

- mucosal ulcers with surrounding translucent mound of edema.
- Earliest radiological and endoscopic finding.



Aphthous ulcers

Rose thorn (Transmural ulcers)

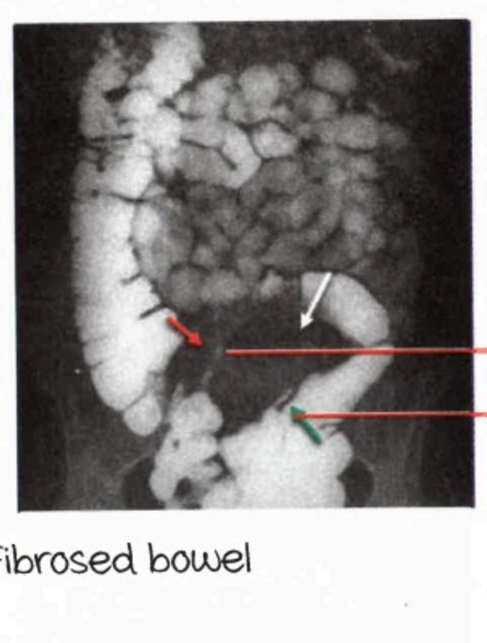


Cobble stone appearance



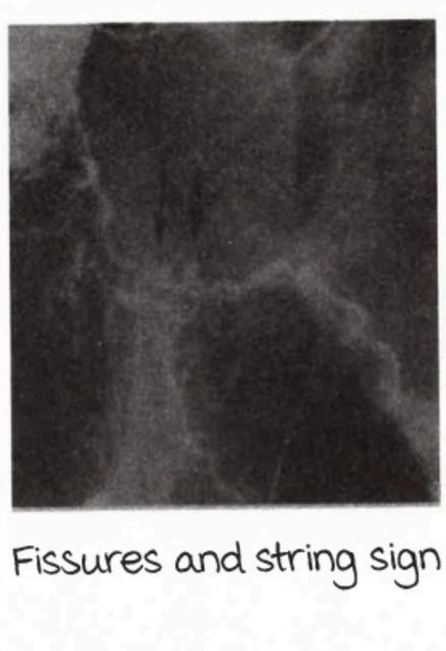


String sign : Fibrosed bowel



String sign

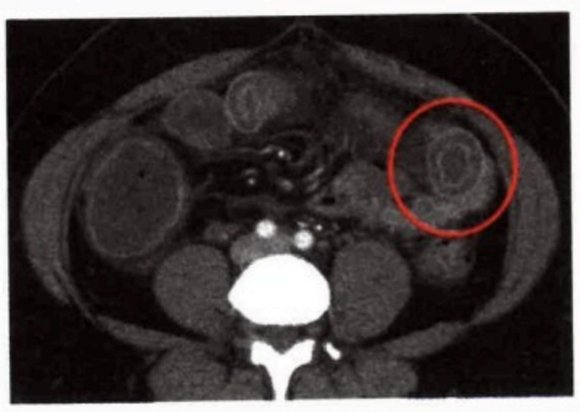
Sinus tract



Fissures and string sign

CT scan :

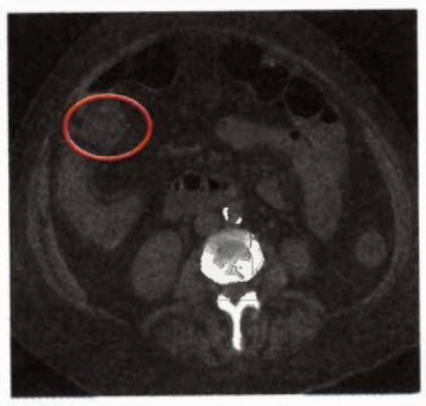
- Limited role in the diagnosis of IBD (MR enterography preferred instead).
- Findings :



Target appearance (Colonic wall thickening)



Comb sign in Crohn's disease (Hypervascular mesentery)



Fat halo sign

- Inner, outer layer : Soft tissue attenuation.
- middle layer : Fatty attenuation.

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CROHN'S DISEASE VS INTESTINAL TUBERCULOSIS

	Crohn's Disease	Intestinal Tuberculosis
Involvement of ileocecal valve (ICV)	Common	Usually strictured and patulous ICV
Involvement of colon	≥4 segments, especially terminal ileum and ICV	<4 segments, especially ICV and ascending colon
Ulcers	Longitudinal	Transverse
Cobblestone appearance	Frequent	Rare
Aphthous lesions	Frequent	Rare
Perianal lesions	Frequent	Rare

Extra-intestinal Manifestations Of IBD

00:23:25

most extra-intestinal manifestations are m/c in Crohn's disease.

m/c in ulcerative colitis :

- Uveitis.
- Primary sclerosing cholangitis.
- Pyoderma gangrenosum.

	Parallels bowel activity	Unrelated to bowel activity
Eye	Episcleritis : Extension of conjunctivitis.	Uveitis : more dangerous (Can lead to loss of vision).
Skin	1. Erythema nodosum  Painful, Pretibial, Papules (PPP) 2. Sweet syndrome. 3. Pyoderma vegetans. 4. Aphthous ulcers.	Pyoderma gangrenosum 
Joints	Peripheral arthritis Type 1 : Large joint migratory asymmetrical pauciarticular arthritis (LMAP).	• Peripheral arthritis Type 2 . • Ankylosing spondylitis like axial arthritis.

OTHERS

1. Primary sclerosing cholangitis (PSC) :
   - m/c in ulcerative colitis.
   - Fibrosing destruction of intra and extrahepatic bile ducts.
   - 2/3rd of patients with PSC : UC +ve.
   - 5% of patients with UC : PSC +ve.

2. Stones :
   - Cholelithiasis.
   - Nephrolithiasis (Oxalate).

3. Musculoskeletal :
   - Osteopenia, osteoporosis.
   - m/c : >40% patients, d/t long term drug use.

4. Liver Steatosis.
5. ↑ risk of thromboembolism.

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Other Investigations

00:31:05

FECAL CALPROTECTIN AND LACTOFERRIN

- Helps to differentiate IBD relapse from infectious large bowel diarrhea.
- Levels : ↑sed in IBD relapse.

Notes : Relapse identified by

a. Fecal calprotectin & lactoferrin.

b. Correlation with extraintestinal manifestations.

SEROLOGY

Ulcerative colitis :

- Atypical p-ANCA : Correlates with disease activity.
- Anti-goblet cell antibody.

Crohn's disease :

- ASCA : Anti saccharomyces cerevisiae antibody.
- OMP-C : Outer membrane porin C antibody.
- APB : Anti-pancreatic antibody.
- Anti-I_2 : Elevated → Bad prognosis, need for surgery.
- Anti-flagellin.

Note :

ASCA is Common to Crohn's disease and Behçet's disease.

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Treatment Of Ulcerative Colitis

00:37:45

TRUELOVE & WITT'S CLASSIFICATION

Mild :
- <4 stools/day
- No fever
- No tachycardia
- Mild anemia
- ESR <30 mm/hr

Moderate :
Between mild and Severe

Severe :
- >6 stools/day with blood
- Fever >37.5°C
- HR >90 bpm
- Anemia with Hb level <75% of normal
- ESR >30mm/hr

MEDICAL MANAGEMENT

- 5-ASA (Aminosalicylic acid) :
  - Main stay Rx.
  - Disadvantage : Absorbed at small intestine → Does not reach rectum (site of active disease).
- Mesalamine :
  - Delayed/controlled release 5-ASA.
  - DOC for mild UC.

Management

	Mild UC :	Moderate UC :	Severe UC :
Induction : therapy	Topical 5-ASA/Oral mesalamine	• Oral mesalamine + Oral steroids • Steroid resistant/refractory : Azathioprine/6-mercaptopurine.	IV steroid + I.V anti-TNF-α /I.V Cyclosporine
maintenance :	Topical 5-ASA	• Topical 5-ASA + Azathioprine/6-mercaptopurine • No role of steroids	Azathioprine/6-mercaptopurine + Low dose anti-TNF-α/Oral cyclosporine

Note :

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- Sulfasalazine :
  - 5-ASA + Sulfapyridine (Carrier) → Absorbed at rectum.
  - S/E (D/t sulfapyridine) : Hypersensitivity, bone marrow suppression, G6PD deficiency → No longer used.
- S/E of azathioprine : Hepatotoxicity, bone marrow toxicity, pancreatitis.

SURGICAL MANAGEMENT

Procto-colectomy with ileoanal anastomosis.

Indications :

- Colonic dysplasia or carcinoma.
- Uncontrollable colonic hemorrhage.
- Colonic perforation.
- Toxic megacolon.
- medically refractory disease.

Crohn's Disease

00:45:00

GRADING OF SEVERITY

mild - moderate :
- No dehydration
- No obstruction
- No weight loss
- No systemic features

moderate - Severe :
Between mild and severe

Severe - Fulminant :
Despite steroid intake, patient has :
- Fever
- Cachexia
- Obstruction
- Peritoneal signs

MEDICAL MANAGEMENT

Objectives :

- Endoscopic mucosal healing.
- Prevent complications (Strictures, fistulae etc).

Drugs :

- Antibiotics.
- Corticosteroids (High risk of dependency and / or resistance).
- 5-ASA.
- Immunomodulators.
- Biologics.
  - Anti TNF agents : Infliximab, Adalimumab, Certolizumab pegol.
  - Play an important role in the treatment of continuously active disease/prolonged symptomatic remission.

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Note : Avoid smoking and NSAIDs.

Management

	Mild – Moderate :	Moderate – Severe :	Severe – Fulminant :
Induction therapy :	Oral Budesonide 9mg/day	Oral steroids + methotrexate + Anti – TNF α	IV Steroids + IV anti TNF-α
Maintenance :	Not required	Azathioprine/6-mercaptopurine + methotrexate + Periodic anti – TNF α	Periodic anti-TNF-α

Fistulizing disease :

Anti-TNFα ± Immunomodulators followed by surgery.

Note :

- Drugs to be avoided in maintenance therapy :
  - Oral 5-ASA.
  - Corticosteroids.
  - Budesonide.
- Newer drugs :
  - Natalizumab : Anti-α4 integrin.
  - Ustekinumab : Anti-IL 12/23.
  - Vedolizumab : Anti-integrin.

Malignancy In IBD

00:56:46 ----- Active space -----

Risk of malignancy is present in both UC and CD.

	Crohn's Disease	Ulcerative Colitis
↑ risk of :	• Non Hodgkin's lymphoma. • Leukemia. • myelodysplastic syndrome.	Cholangiocarcinoma (D/t association with primary sclerosing cholangitis)

SPORADIC VS IBD-RELATED COLORECTAL CARCINOMA

	Sporadic Colorectal Carcinoma	IBD related carcinoma
Age of onset	>50yrs	10-15yrs younger
5-yr survival rate	50% for both	
Genetic sequence	Adenoma-carcinoma sequence	Inflammation-Dysplasia-Carcinoma sequence
Site of lesion	Common on left side	Any segment of diseased colon
Pathological feature	Adenocarcinoma	Signet ring cell carcinoma
Genetics	P53 mutation : Late APC mutation : Early	P53 mutation : Early APC mutation : Late